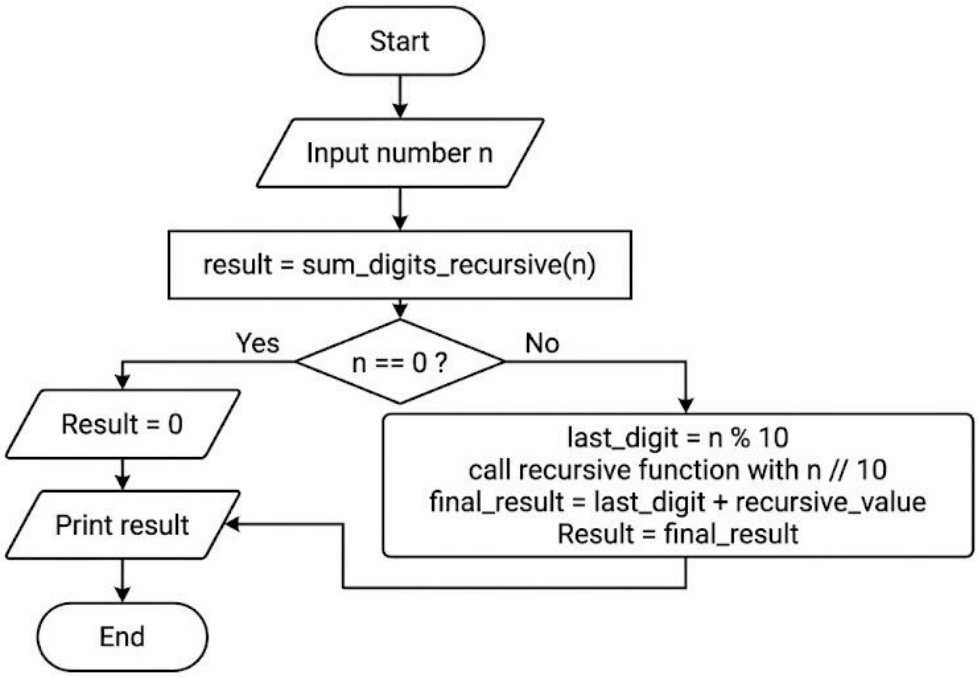
## 11.1.2 Sum of Digits using Recursion

## Algorithm

Step 1: Start
Step 2: Input number n
Step 3: Call function sum_of_digits_recursive(n)
Step 4: Check if n == 0

- If YES → return 0
- If NO → go to next step
  Step 5: Calculate (n % 10) (last digit)
  Step 6: Call function again with n // 10
  Step 7: Add result → (n % 10) + recursive call
  Step 8: Return result
  Step 9: Print result
  Step 10: Stop

## Flowchart



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## 11.1.2. Sum of Digits using Recursion

02:28

Write a recursive function **sum_of_digits_recursive(n)** that calculates the sum of the digits of a given non-negative integer $n$.

**Input Format:**

- The input consists of a single non-negative integer $n$.

**Output Format:**

- The output should be the sum of the digits of the non-negative integer $n$.

**Constraints:**

- $0 \leq n \leq 10^9$

Sample Test Cases +

Explorer

SumofDi...

Submit

Debugger

```python
def sum_of_digits_recursive(n):
    if n==0:
        return 0
    return n%10+ sum_of_digits_recursive( n//10 )
number=int(input())
result = sum_of_digits_recursive(number)
print(result)
```

| Average time | Maximum time |
|---|---|
| 0.019 s | 0.029 s |
| 19.40 ms | 29.00 ms |

2 out of 2 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 20 ms Debug

| Expected output | Actual output |
|---|---|
| 23 | 23 |
| 5 | 5 |

Test case 2 29 ms

Terminal Test cases

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